

Optimizer Performance Comparison (2 Epochs)

Global Summary

Training Loss and Final Accuracy

Optimizer	Epoch 1 Loss	Epoch 2 Loss	Final Accuracy
SGD	2.5918	1.7167	58.33%
Momentum	1.3575	0.3501	92.78%
Adagrad	1.1889	0.4636	90.83%
RMSprop	1.8771	0.8751	85.56%
Adadelta	1.2097	0.2919	96.11%
Adam	2.1408	1.0186	80.56%
NAdam	1.8689	0.8565	83.89%

1. SGD (Stochastic Gradient Descent)

SGD - Classification Report

Class	Precision	Recall	F1-Score	Support
0	0.64	0.85	0.73	33
1	0.50	0.61	0.55	28
2	0.69	0.67	0.68	33
3	0.50	0.56	0.53	34
4	0.61	0.85	0.71	46
5	0.72	0.28	0.40	47
6	0.71	0.77	0.74	35
7	0.52	0.68	0.59	34
8	0.40	0.07	0.11	30
9	0.47	0.50	0.48	40
Accuracy		0.58		360
Macro Avg	0.58	0.58	0.55	360
Weighted Avg	0.58	0.58	0.55	360

SGD - Confusion Matrix

		Predicted Class									
		0	1	2	3	4	5	6	7	8	9
True Class	0	28	1	2	0	1	0	0	1	0	0
	1	0	17	3	5	0	0	0	0	2	1
	2	1	2	22	1	3	1	2	0	1	0
	3	2	3	4	19	0	1	2	1	0	2
	4	2	0	0	1	39	1	2	1	0	0
	5	2	0	0	2	3	13	3	10	0	14
	6	6	0	0	0	0	0	27	2	0	0
	7	0	0	1	0	8	1	0	23	0	1
	8	0	8	0	4	6	1	2	2	2	5
	9	3	3	0	6	4	0	0	4	0	20

2. Momentum

Momentum - Classification Report

Class	Precision	Recall	F1-Score	Support
0	0.97	0.94	0.95	33
1	0.81	0.89	0.85	28
2	0.94	0.97	0.96	33
3	1.00	0.94	0.97	34
4	0.98	1.00	0.99	46
5	0.95	0.85	0.90	47
6	0.94	0.97	0.96	35
7	0.94	0.97	0.96	34
8	0.96	0.80	0.87	30
9	0.80	0.93	0.86	40
Accuracy		0.93		360
Macro Avg	0.93	0.93	0.93	360
Weighted Avg	0.93	0.93	0.93	360

Momentum - Confusion Matrix

		Predicted Class									
		0	1	2	3	4	5	6	7	8	9
True Class	0	31	0	0	0	1	0	1	0	0	0
	1	0	25	1	0	0	0	0	0	0	2
	2	0	1	32	0	0	0	0	0	0	0
	3	0	1	1	32	0	0	0	0	0	0
	4	0	0	0	0	46	0	0	0	0	0
	5	0	1	0	0	0	40	1	0	0	5
	6	1	0	0	0	0	0	34	0	0	0
	7	0	0	0	0	0	0	0	33	0	1
	8	0	3	0	0	0	1	0	1	24	1
	9	0	0	0	0	0	1	0	1	1	37

3. Adagrad

Adagrad - Classification Report

Class	Precision	Recall	F1-Score	Support
0	0.94	0.94	0.94	33
1	0.90	0.93	0.91	28
2	0.94	0.94	0.94	33
3	0.88	0.85	0.87	34
4	0.98	0.98	0.98	46
5	0.93	0.91	0.92	47
6	0.97	0.97	0.97	35
7	0.92	0.97	0.94	34
8	0.81	0.83	0.82	30
9	0.79	0.75	0.77	40
Accuracy		0.91		360
Macro Avg	0.91	0.91	0.91	360
Weighted Avg	0.91	0.91	0.91	360

Adagrad - Confusion Matrix

		Predicted Class									
		0	1	2	3	4	5	6	7	8	9
True Class	0	31	0	1	0	0	0	0	1	0	0
	1	0	26	0	0	0	0	0	0	0	2
	2	0	0	31	1	0	0	0	0	1	0
	3	0	0	1	29	0	1	0	0	2	1
	4	0	1	0	0	45	0	0	0	0	0
	5	0	0	0	0	0	43	1	0	0	3
	6	1	0	0	0	0	0	34	0	0	0
	7	0	0	0	0	0	0	0	33	0	1
	8	0	2	0	0	1	1	0	0	25	1
	9	1	0	0	3	0	1	0	2	3	30

4. RMSprop

RMSprop - Classification Report

Class	Precision	Recall	F1-Score	Support
0	0.94	0.91	0.92	33
1	0.68	0.82	0.74	28
2	0.72	0.88	0.79	33
3	0.86	0.94	0.90	34
4	0.94	1.00	0.97	46
5	0.87	0.85	0.86	47
6	0.94	0.97	0.96	35
7	0.91	0.91	0.91	34
8	0.92	0.37	0.52	30
9	0.80	0.80	0.80	40
Accuracy		0.86		360
Macro Avg	0.86	0.85	0.84	360
Weighted Avg	0.86	0.86	0.85	360

RMSprop - Confusion Matrix

		Predicted Class									
		0	1	2	3	4	5	6	7	8	9
True Class	0	30	1	0	0	0	1	1	0	0	0
	1	0	23	4	0	0	0	0	0	0	1
	2	0	2	29	1	0	0	0	0	1	0
	3	0	0	1	32	0	0	0	0	0	1
	4	0	0	0	0	46	0	0	0	0	0
	5	0	0	0	0	0	40	1	0	0	6
	6	1	0	0	0	0	0	34	0	0	0
	7	0	0	0	0	2	1	0	31	0	0
	8	0	7	6	3	0	2	0	1	11	0
	9	1	1	0	1	1	2	0	2	0	32

5. Adadelata

Adadelata - Classification Report

Class	Precision	Recall	F1-Score	Support
0	0.97	1.00	0.99	33
1	0.96	0.93	0.95	28
2	0.94	1.00	0.97	33
3	1.00	0.94	0.97	34
4	1.00	1.00	1.00	46
5	0.96	0.94	0.95	47
6	0.97	0.97	0.97	35
7	1.00	0.97	0.99	34
8	0.90	0.93	0.92	30
9	0.90	0.93	0.91	40
Accuracy		0.96		360
Macro Avg	0.96	0.96	0.96	360
Weighted Avg	0.96	0.96	0.96	360

Adadelata - Confusion Matrix

		Predicted Class									
True Class		0	1	2	3	4	5	6	7	8	9
	0	33	0	0	0	0	0	0	0	0	0
	1	0	26	1	0	0	0	0	0	0	1
	2	0	0	33	0	0	0	0	0	0	0
	3	0	0	1	32	0	0	0	0	1	0
	4	0	0	0	0	46	0	0	0	0	0
	5	0	0	0	0	0	44	1	0	0	2
	6	1	0	0	0	0	0	34	0	0	0
	7	0	0	0	0	0	0	0	33	0	1
	8	0	1	0	0	0	1	0	0	28	0
	9	0	0	0	0	0	1	0	0	2	37

6. Adam

Adam - Classification Report

Class	Precision	Recall	F1-Score	Support
0	0.85	0.88	0.87	33
1	0.63	0.79	0.70	28
2	0.81	0.76	0.78	33
3	0.82	0.79	0.81	34
4	0.94	0.98	0.96	46
5	0.84	0.81	0.83	47
6	0.85	0.97	0.91	35
7	0.86	0.91	0.89	34
8	0.81	0.43	0.57	30
9	0.62	0.65	0.63	40
Accuracy		0.81		360
Macro Avg	0.80	0.80	0.79	360
Weighted Avg	0.81	0.81	0.80	360

Adam - Confusion Matrix

		Predicted Class									
		0	1	2	3	4	5	6	7	8	9
True Class	0	29	0	0	0	2	0	2	0	0	0
	1	0	22	2	0	0	1	1	0	0	2
	2	0	3	25	0	0	1	0	2	1	1
	3	0	1	0	27	0	0	0	1	2	3
	4	0	0	0	0	45	0	0	0	0	1
	5	3	1	1	0	0	38	1	0	0	3
	6	1	0	0	0	0	0	34	0	0	0
	7	0	0	2	0	0	0	0	31	0	1
	8	0	5	1	1	0	1	2	2	13	5
	9	1	3	0	5	1	4	0	0	0	26

7. NAdam

NAdam - Classification Report

Class	Precision	Recall	F1-Score	Support
0	0.94	0.94	0.94	33
1	0.72	0.82	0.77	28
2	0.82	0.85	0.84	33
3	0.80	0.94	0.86	34
4	0.95	0.89	0.92	46
5	0.86	0.68	0.76	47
6	0.87	0.97	0.92	35
7	0.83	0.88	0.86	34
8	0.90	0.63	0.75	30
9	0.71	0.80	0.75	40
Accuracy		0.84		360
Macro Avg	0.84	0.84	0.84	360
Weighted Avg	0.85	0.84	0.84	360

NAdam - Confusion Matrix

		Predicted Class									
		0	1	2	3	4	5	6	7	8	9
True Class	0	31	0	0	0	1	1	0	0	0	0
	1	0	23	3	1	0	0	0	0	1	0
	2	0	1	28	2	0	0	0	2	0	0
	3	0	1	0	32	0	0	0	0	1	0
	4	0	2	0	0	41	0	3	0	0	0
	5	0	1	1	0	0	32	1	1	0	11
	6	1	0	0	0	0	0	34	0	0	0
	7	0	0	0	2	0	2	0	30	0	0
	8	1	3	2	0	0	0	1	2	19	2
	9	0	1	0	3	1	2	0	1	0	32